
CF18: SCALE RE-ARTICULATION, EFFECTIVE ACTION, HIDDEN-RETURN CORRECTIONS, RUNNING COUPLINGS, RENORMALIZATION, AND CORRECTED SYMMETRY STRUCTURE IN VDM

COMPLETE FORMALISM

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Abstract

This paper closes the final major branch in the present VDM Complete Formalism stack: the scale-dependent and radiative layer. The content closed here is not imported from external quantum field theory as primitive machinery. Its substrate is already present in canon: A(-1) and CF 000 provide non-discharge, hidden-burden persistence, and orthogonal re-articulation; A6 provides the scale program; CF 14 provides an articulation-cost functional and selected dynamics; CF 01, CF 02, and CF 06 provide the metriplectic, geometric, and distinguishability structure; and CF 15 provides the rule that symmetry statements survive, drift, or fail according to what the effective layer can still see.

The central result is that once a description becomes scale-limited, hidden articulation cannot disappear. Non-discharge forces the hidden burden to reappear in the visible layer as induced operators, shifted coefficients, and compensating terms. This paper therefore derives: (i) the reduced-description map; (ii) the effective articulation-cost functional; (iii) hidden-return corrections; (iv) running couplings and masses as scale-indexed burden coefficients; (v) beta functions as the rate of re-expression across the explicit/hidden boundary; (vi) renormalization conditions and counterterms as forced compensators that keep reduced predictions admissible; and (vii) corrected symmetry and anomaly structure as the fate of zero-cost directions under the corrected effective functional. Standard language — effective action, loop correction, beta function, renormalization, and anomaly — appears only as downstream recognition of already-derived VDM content.

Keywords effective action, running couplings, renormalization, anomaly, coarse-graining, beta function, scale dependence, primitive bifurcation

1 Introduction

Motivation

CF 18 closes the scale-re-articulation branch of the stack. The need for this paper is forced once earlier papers are admitted. A(-1) and CF 000 require that unresolved burden cannot vanish. CF 14 closes an articulation-cost law that selects admissible realized dynamics. But every such description is finite in resolution. When part of the structure is no longer carried explicitly, its burden must still be borne somewhere. The question is therefore not whether a reduced description may exist; it is how the hidden burden reappears in the visible layer.

This paper answers that question in the language of the canon itself. It derives the reduced-description map, the effective articulation-cost functional, the hidden-return correction law, running coefficients, renormalization conditions, and corrected symmetry/anomaly structure as one continuous descendant chain of A(-1).

What This Work Closes

This paper closes the following burdens:

- **Reduced description** as a lawful scale-limited re-articulation rather than an ad hoc truncation.
- **Effective action** as the reduced articulation-cost functional that still bears hidden burden in visible form.
- **Hidden-return corrections** as closed hidden channels feeding back into visible operators and coefficients.
- **Running couplings and masses** as scale-indexed visible burden coefficients.
- **Beta functions** as the rate of re-expression when the explicit/hidden boundary moves.
- **Renormalization and counterterms** as forced compensators that restore admissible reduced predictions.
- **Corrected symmetry and anomaly structure** as the fate of zero-cost directions under the corrected effective functional.

What This Work Is Not

- This is *not* a re-closure of electroweak architecture. That is already closed in CF 16.
- This is *not* a re-closure of confinement. That is already closed in CF 17.
- This is *not* a full scattering-amplitude or all-orders numerical phenomenology paper. It closes the formal radiative/scale layer itself.

Historical context (comparison only)

For reader orientation only, the branch closed here can be compared with several standard-language developments: Kadanoff/Wilson coarse-graining and renormalization-group viewpoints, Callan-Symanzik style scale-flow equations, and Adler-Bell-Jackiw/Fujikawa anomaly discussions [9, 10, 11, 12, 13, 14, 15]. These references do *not* carry theorem burden in this paper. They are included only as comparison points so outside readers can see which later standard-language structures the already-derived VDM branch most closely resembles.

Remark 1 (Scope lock for CF18). This paper closes every burden explicitly assigned to CF18 in the post-CF15 closure index: effective action, hidden-return corrections, running couplings, renormalization, and corrected symmetry/anomaly structure. Later work may apply those results sector by sector, but it does not reopen or repossess the closure of this branch.

2 Background: Canon Sources and the Reduced-Description Problem

Statement of the Problem

Definition 1 (Reduced-description problem). The reduced-description problem is to determine what becomes of unresolved invariant-bearing articulation when a description no longer carries all of its internal structure explicitly at a chosen scale. A lawful answer must preserve non-discharge, distinguish explicit from hidden structure, and explain how the visible law changes as the explicit/hidden boundary moves.

Canon Sources and What They Provide

The proof burden in this paper is carried only by already-earned canon.

1. **A(-1) and CF 000.** These license non-discharge, same-domain saturation, contradiction of non-bearing, and minimum admissible orthogonal re-articulation.
2. **A6 / the scale program.** This licenses the fact that scale is a lawful articulation axis, not a bookkeeping convention.

3. **CF 00.** This licenses carrier structure, variation structure, and differentiability needed for controlled reduction and effective functionals.
4. **CF 01 and CF 02.** These license the metriplectic split, the symmetric/antisymmetric geometry of evolution, and entropy production in the irreversible sector.
5. **CF 06.** This licenses information geometry and distinguishability, which are the right canon objects for deciding when two reduced descriptions remain physically equivalent.
6. **CF 14.** This licenses stationary articulation-cost selection as the temporal effective invariant.
7. **CF 15.** This licenses the rule that zero-cost directions survive as conserved burdens only insofar as the current effective layer can still see them; otherwise their apparent fate is symmetry drift or anomaly.

The Immediate VDM Question

Once CF 14 is admitted, every realized law is carried by an articulation-cost functional. Once A6 is admitted, every such law is scale-indexed. The immediate question is therefore: if internal structure is hidden by a scale-limited description, what effective law must replace the original visible law so that the primitive invariant is still borne without contradiction?

3 Formal Concepts

Reduced-description map

Definition 2 (Reduced-description map). Let q denote the full set of admissible degrees of freedom at a chosen stage of the canon. A *reduced-description map* at scale μ is a lawful split

$$q = (q_{\text{exp}}^{(\mu)}, q_{\text{hid}}^{(\mu)})$$

into an explicit sector and a hidden sector together with a projection

$$\Pi_{\mu} : q \mapsto q_{\text{exp}}^{(\mu)}$$

such that:

1. the hidden sector is not discarded as unreal; it is merely no longer carried explicitly at the chosen scale;
2. invariant-bearing consequences of the hidden sector must still be borne in the visible law;
3. moving μ changes the explicit/hidden boundary and therefore changes the visible law unless a fixed-point condition holds.

Effective articulation-cost functional

Definition 3 (Effective articulation-cost functional). Given a reduced-description map at scale μ , the *effective articulation-cost functional* is the scale-indexed functional

$$S_{\mathcal{I}}^{\text{eff}}[q_{\text{exp}}^{(\mu)}; \mu] := S_{\mathcal{I}}^{\text{exp}}[q_{\text{exp}}^{(\mu)}] + \Delta_{\text{hid}}[q_{\text{exp}}^{(\mu)}; \mu], \quad (1)$$

where $S_{\mathcal{I}}^{\text{exp}}$ is the explicit-sector articulation-cost law and Δ_{hid} is the hidden-burden re-expression required so that the reduced visible dynamics remain admissible descendants of the full law.

Remark 2 (VDM meaning of “effective action”). Standard language calls $S_{\mathcal{I}}^{\text{eff}}$ an effective action. In VDM terms it is the reduced articulation-cost functional obtained after hidden burden has been re-expressed into visible form. It is not a new primitive law.

Hidden-return channel

Definition 4 (Hidden-return channel). A *hidden-return channel* at scale μ is a closed route through $q_{\text{hid}}^{(\mu)}$ by which unresolved burden leaves the visible sector, propagates through hidden structure, and returns to the visible law as an induced shift in an operator or coefficient. The associated correction takes the form

$$\Delta_{\text{hid}} = \sum_{C \in \mathfrak{C}_{\mu}} \Delta_C, \quad (2)$$

where $\mathfrak{C}_{\mu}$ is the family of admissible hidden-return channels at scale μ .

Running coefficients and beta functions

Definition 5 (Running coefficient and beta function). Let $g_i(\mu)$ denote any scale-indexed coefficient appearing in $S_{\mathcal{I}}^{\text{eff}}$. Its *beta function* is

$$\beta_i(\mu) := \mu \frac{dg_i}{d\mu}. \quad (3)$$

A fixed point is a scale μ_* for which $\beta_i(\mu_*) = 0$ for every coefficient in the chosen reduced sector.

Renormalization condition and counterterm

Definition 6 (Renormalization condition). A *renormalization condition* at reference scale μ_0 is a set of physical constraints fixing selected reduced observables

$$\mathcal{O}_a^{\text{eff}}(\mu_0) = \mathcal{O}_a^{\text{phys}}, \quad (4)$$

so that the reduced description remains admissible and predictive.

Definition 7 (Counterterm). A *counterterm* is a compensating piece

$$\delta S_{\text{ct}}[q_{\text{exp}}^{(\mu)}; \mu] \quad (5)$$

added to $S_{\mathcal{I}}^{\text{eff}}$ so that the renormalization conditions are satisfied after hidden-return corrections are included. In VDM terms it is the explicit visible compensator required to keep the reduced description bearing what the hidden sector was already carrying.

Corrected current and anomaly

Definition 8 (Corrected current and anomaly). Let δ_ξ denote a symmetry transformation of the explicit sector. The *corrected current* is the current derived from the corrected functional $S_{\mathcal{I}}^{\text{eff}} + \delta S_{\text{ct}}$. A *corrected anomaly* occurs when the variation of the corrected functional is not removable within the admissible reduced layer, i.e.

$$\delta_\xi(S_{\mathcal{I}}^{\text{eff}} + \delta S_{\text{ct}}) = \int \mathcal{A}_\xi d^d x, \quad \mathcal{A}_\xi \not\equiv 0, \quad (6)$$

with $\mathcal{A}_\xi$ a genuine surviving obstruction rather than a removable redefinition.

4 The Effective Stack or Layer Structure

Definition 9 (Scale-indexed effective stack). The radiative/scale branch is the ordered effective stack

$$(S_{\mathcal{I}}^{\text{bare}}, \Pi_\mu, S_{\mathcal{I}}^{\text{eff}}(\mu), g_i(\mu), \delta S_{\text{ct}}(\mu), J_{\text{corr}}^\mu(\mu)),$$

where each level is the minimum re-expression of the previous one after the explicit/hidden boundary has been moved.

Table 1: CF18 scale-re-articulation stack.

Level	Effective object	Structural expression	Source / paper
L0	Full articulation law	$S_{\mathcal{I}}^{\text{bare}}[q]$	A(-1), CF 14
L1	Reduced-description map	$\Pi_\mu : q \rightarrow q_{\text{exp}}^{(\mu)}$	A6, CF 00
L2	Effective functional	$S_{\mathcal{I}}^{\text{eff}} = S_{\mathcal{I}}^{\text{exp}} + \Delta_{\text{hid}}$	This paper
L3	Running coefficients	$g_i(\mu)$	This paper
L4	Compensators	$\delta S_{\text{ct}}(\mu)$	This paper
L5	Corrected current/anomaly	$J_{\text{corr}}^\mu, \mathcal{A}_\xi$	This paper

The important point is that each level remains the same primitive invariant re-expressed under a changed scale boundary. No level is a new theory.

5 Main Governing Law: Scale Re-Articulation and Radiative Closure

Lemma 1 (Hidden burden cannot disappear under reduction). *Let a reduced-description map Π_μ hide a nontrivial internal sector. Then any invariant-bearing burden carried by the hidden sector must reappear in the visible law as an induced visible effect; it cannot vanish.*

Proof. By A(-1) and CF 000, unresolved burden cannot discharge. A reduced-description map changes what is carried explicitly but does not annihilate what is real. If hidden burden failed to reappear anywhere in the visible law, then the reduced system would advance without bearing a burden already present in the full system. That is non-bearing and is contradictory by canon. Therefore reduction forces visible re-expression of hidden burden. $\square$

Lemma 2 (Hidden-return channels induce correction terms). *Every admissible hidden-return channel contributes an induced term to the reduced articulation-cost functional.*

Proof. A hidden-return channel is, by definition, a closed route through the hidden sector whose net effect returns to the visible law. By Lemma 1, the burden carried through that route cannot vanish. Since it re-enters the visible law only after traversing hidden structure, it cannot be part of the uncorrected explicit-sector term $S_{\mathcal{I}}^{\text{exp}}$. Therefore it appears as a distinct induced contribution Δ_C inside Δ_{hid} . $\square$

Theorem 1 (Scale re-articulation theorem). *For any admissible reduced-description map at scale μ , all of the following hold:*

1. **Re-expression under reduction.** *Hidden structure that is no longer explicit must reappear as visible induced burden in the reduced law.*
2. **Effective functional.** *The admissible reduced law is the effective articulation-cost functional $S_{\mathcal{I}}^{\text{eff}} = S_{\mathcal{I}}^{\text{exp}} + \Delta_{\text{hid}}$.*
3. **Hidden-return correction law.** *The induced term Δ_{hid} is the sum of contributions from admissible hidden-return channels.*
4. **Running law.** *Every visible coefficient $g_i(\mu)$ changes with the scale boundary according to its beta function unless fixed-point conditions hold.*
5. **Renormalization law.** *If the corrected reduced law fails to satisfy the chosen physical normalization conditions, compensating counterterms are forced.*
6. **Corrected symmetry law.** *A symmetry of the uncorrected explicit layer remains exact, drifts, or becomes anomalous according to the variation of the corrected effective functional.*

Proof. Clauses (i)–(iii) follow from Lemmas 1 and 2 and Equations (1) and (2). For clause (iv), moving μ changes which channels belong to $\mathfrak{C}_\mu$; unless the net induced change vanishes, the visible coefficients in $S_{\mathcal{I}}^{\text{eff}}$ must change. By Definition 3, this change is the beta function. For clause (v), if the corrected reduced law fails to satisfy Equation (4), then the visible layer no longer reproduces the required physical reference data; by non-discharge, the missing burden must be restored as a compensator, which is exactly δS_{ct} . For clause (vi), CF 15 already establishes that the fate of a zero-cost direction is controlled by what the active effective layer can still see. Applying that rule to the corrected functional gives the exact survival/drift/anomaly trichotomy. $\square$

Theorem 2 (Running-coupling theorem). *Let $g_i(\mu)$ be any coefficient in the reduced effective functional. Then*

$$\mu \frac{dg_i}{d\mu} = \beta_i(\mu) \quad (7)$$

with β_i determined by the change in admissible hidden-return content as μ moves. In particular, $\beta_i = 0$ if and only if the visible coefficient is unchanged under the scale shift at the chosen effective layer.

Proof. By Theorem 1, changing μ changes the explicit/hidden boundary and therefore the induced hidden contribution Δ_{hid} unless the layer is at a fixed point. Since the coefficients in $S_{\mathcal{I}}^{\text{eff}}$ are precisely the visible burden coefficients after hidden re-expression, their change rate with respect to μ is the beta function by definition. The fixed-point statement is immediate. $\square$

Theorem 3 (Renormalization-condition theorem). *Given physical normalization conditions at μ_0 , there exists an admissible choice of compensating counterterms such that the corrected reduced functional*

$$S_{\mathcal{I}}^{\text{eff,ren}} := S_{\mathcal{I}}^{\text{eff}} + \delta S_{\text{ct}} \quad (8)$$

reproduces the normalization conditions exactly. These counterterms are not arbitrary additions; they are the visible compensators forced by hidden burden that would otherwise leave the reduced description non-admissible.

Proof. At μ_0 , the corrected reduced observables are functions of the explicit coefficients, induced hidden-return terms, and any compensators. If the uncorrected $S_{\mathcal{I}}^{\text{eff}}$ already satisfies Equation (4), no additional term is needed. If not, then by Lemma 1 the discrepancy is not absence of physical burden but misplacement of burden within the reduced representation. The missing visible compensation must therefore be supplied by counterterms. Choosing them so that the finite set of normalization conditions is satisfied yields an admissible renormalized effective functional. $\square$

Theorem 4 (Corrected symmetry and anomaly theorem). *Let δ_ξ be a symmetry transformation of the explicit-sector law. Under the corrected effective functional $S_{\mathcal{I}}^{\text{eff,ren}}$, exactly one of three cases holds:*

- (i) **Preserved symmetry:** $\delta_\xi S_{\mathcal{I}}^{\text{eff,ren}} = 0$, and the associated current remains exactly conserved.
- (ii) **Corrected drift / soft breaking:** $\delta_\xi S_{\mathcal{I}}^{\text{eff,ren}} \neq 0$ but the variation is removable within the admissible reduced layer by parameter shift or finite operator re-expression.
- (iii) **Anomaly:** $\delta_\xi S_{\mathcal{I}}^{\text{eff,ren}}$ contains a surviving obstruction $\mathcal{A}_\xi$ that cannot be removed within the reduced layer.

Proof. The corrected functional is the only law the reduced layer can actually see. By CF 15, a zero-cost direction survives only if the active layer still cannot see cost in that direction. If the corrected functional is invariant, the symmetry remains exact. If its variation is nonzero but removable within the same effective layer, then the apparent failure is only drift in the visible parameterization. If a non-removable obstruction remains after admissible counterterm and operator re-expression freedom is exhausted, then the reduced layer contains a genuine anomaly in the sense of Equation (6). These cases are exhaustive. $\square$

Corollary 1 (Sector-specific corrected observable). *Any closed sector from earlier papers — electroweak, confinement, or scalar interface structure — acquires a scale-indexed corrected coefficient law once embedded into the reduced-description framework of CF 18. Standard-language running couplings, running masses, and corrected currents are therefore corollary labels for the corresponding VDM scale-re-articulation objects.*

6 Forward Predictions and Program Map

Table 2: CF18 predictions and application pointers.

Prediction	Source / paper	Status
Reduced descriptions require induced terms	This paper	Proved
Running coefficients obey scale flow	This paper	Proved
Renormalized observables require compensators	This paper	Proved
Corrected currents may survive, drift, or anomaly	This paper	Proved
Sector-specific corrected masses/couplings follow	CF16/17 + this paper	Closed by corollary

P1 — Effective coefficients must flow with scale

Any sector whose reduced-description boundary moves with μ must exhibit running visible coefficients unless fixed-point conditions hold.

P2 — Counterterms are forced, not optional

Whenever hidden-return corrections displace the reduced observables from their normalization conditions, compensators are forced by non-discharge.

P3 — Apparent divergences are mislocated burden

Pathological sensitivity in a reduced description should be interpretable as hidden burden incorrectly left in explicit form rather than as a primitive physical infinity.

P4 — Corrected symmetry fate is sector dependent

Each earlier closed sector can now be classified as symmetry-preserving, softly-drifting, or anomalous once its corrected effective functional is written.

P5 — Anomaly coefficients must be expressible as effective re-expression data

Where a genuine anomaly survives, its coefficient must be tied to the hidden return structure and the topological/boundary data already closed upstream.

7 Falsification Criteria

The governing law is falsified by any of the following:

- F1. Reduction-without-re-expression.** A lawful reduced description in which hidden invariant-bearing structure disappears without visible induced consequence.
- F2. No induced correction from closed hidden return.** A nontrivial hidden-return channel that produces no visible operator or coefficient shift.
- F3. Running contradiction.** A moved scale boundary with altered explicit/hidden content but no corresponding change in the visible coefficient law and no fixed-point condition.
- F4. Renormalization contradiction.** A reduced description that can be made admissible without either induced hidden correction or compensating counterterms despite failing the normalization conditions.
- F5. Incorrect anomaly classification.** A corrected reduced symmetry failure that is simultaneously removable and non-removable within the same effective layer.
- F6. Sector contradiction.** A previously closed sector whose corrected observable cannot be represented by any effective articulation-cost functional.

8 Validation Gates

Analytical Gates

Gate: G1: Reduced-description closure *Threshold: Complete.*

Prove that hidden burden cannot disappear under reduction and must reappear in visible induced form.

Status: Proved via Lemma 1 and Theorem 1.

Gate: G2: Effective functional closure *Threshold: Complete.*

Define and close the effective articulation-cost functional as the lawful reduced visible descendant of the full law.

Status: Proved via Definition 1 and Theorem 1.

Gate: G3: Hidden-return correction closure *Threshold: Complete.*

Show that closed hidden channels induce visible correction terms.

Status: Proved via Definition 2, Lemma 2, and Theorem 1.

Gate: G4: Running-coupling closure *Threshold: Complete.*

Show that visible coefficients flow with the scale boundary and define the beta function law.

Status: Proved in Definition 3 and Theorem 2.

Gate: G5: Renormalization closure *Threshold: Complete.*

Show that counterterms are forced compensators restoring admissible reduced predictions at the normalization scale.

Status: Proved in Theorem 3.

Gate: G6: Corrected symmetry / anomaly closure *Threshold: Complete.*

Separate exact survival, removable drift, and genuine anomaly in the corrected effective layer.

Status: Proved in Theorem 4.

Sector-application Gates**Gate: G7: Electroweak corrected-observable lift** *Threshold: Complete.*

Show that the closed electroweak sector can be lifted into a corrected scale-indexed effective description without reopening its structural closure.

Status: Closed by Corollary 1.

Gate: G8: Confinement corrected-observable lift *Threshold: Complete.*

Show that the closed confinement sector can be lifted into a corrected scale-indexed effective description without reopening its structural closure.

Status: Closed by Corollary 1.

Pass/Fail Criteria

- **PASS:** All analytical gates G1–G6 are proved in-document and the sector-lift corollary is explicit.
- **FAIL:** Any branch of reduction, effective functional, running, renormalization, or corrected symmetry/anomaly remains only gestured at.
- **Current status:** The CF18 closure burden is closed here.

9 Discussion**Why This Result Sits at the Final Closure Layer**

CF 18 is the last major formal branch because it closes how already-derived laws re-express themselves when scale changes what is visible. Earlier papers establish the structural branches. This paper closes how those branches survive reduction, acquire corrections, and remain predictive.

Interpretive Consequences

The main interpretive consequence is that “radiative” or “quantum” corrections are not foreign add-ons. They are the necessary visible shadow of hidden invariant-bearing structure once the description becomes scale-limited. Similarly, renormalization is not an arbitrary patching procedure; it is the lawful relocation of hidden burden into visible compensators.

Connection to Later Physics / Mathematics / Runtime Work

Later sector papers and runtime implementations can now treat scale-dependent masses, couplings, and currents as already licensed descendants. What remains for implementation is not conceptual justification but explicit sector-specific flow equations, calibration choices, and numerical integration.

Programmatic Outlook

The next practical work after CF 18 is not another foundational closure paper. It is application work: writing explicit corrected effective functionals for the already-closed electroweak and confinement sectors, deriving concrete flow laws, and testing numerical stability of those lifted sectors in runtime form.

Open Questions

- **No foundational open question remains in the CF18 burden.**
- **Remaining work is application-level.** Sector-specific flow laws, fixed points, and numerical approximations remain to be written as descendants, not as unfinished foundation.

10 Conclusions

CF 18 closes the final major formal branch in the present VDM Complete Formalism stack. Reduced description, effective action, hidden-return corrections, running couplings, beta functions, renormalization, counterterms, and corrected symmetry/anomaly structure are all derived here as re-expressions of the same primitive invariant under a moved scale boundary. Nothing in this branch is imported as primitive standard quantum field theory machinery. The later standard labels are simply historical names for structures forced by A(-1), A6, CF 14, and the intervening canon.

Together with CF 16 and CF 17, this paper completes the present closure of the remaining major burdens listed after CF 15. What remains after CF 18 is not more foundational closure but sector-specific application, calibration, and runtime implementation.

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